Hazidien Ramadhan Utomo

**CLICK FOR MY PORTOFOLIO WEBSITE VERSION**

# PORTFOLIO

@Hazidien

https://www.linkedin.com/in/hazidienramadhanutomo/

hazidienramadhanutomo@gmail.com

2025

# ABOUT ME

GIS SPECIALIST

## HI I'M HAZIDIEN RAMADHAN UTOMO

Urban and Regional Planning graduate from Institut Teknologi Sepuluh Nopember (ITS) with experience in spatial analysis, GIS, and remote sensing. Applied ArcGIS, ArcPy, and Global Mapper to assess rooftop solar potential across approximately 786,527 buildings in 31 districts of Surabaya, integrating LiDAR-derived data and solar radiation modelling. Experienced in land-use change modelling and transportation planning, with additional strengths in analytical problem-solving, team leadership, and coordination.

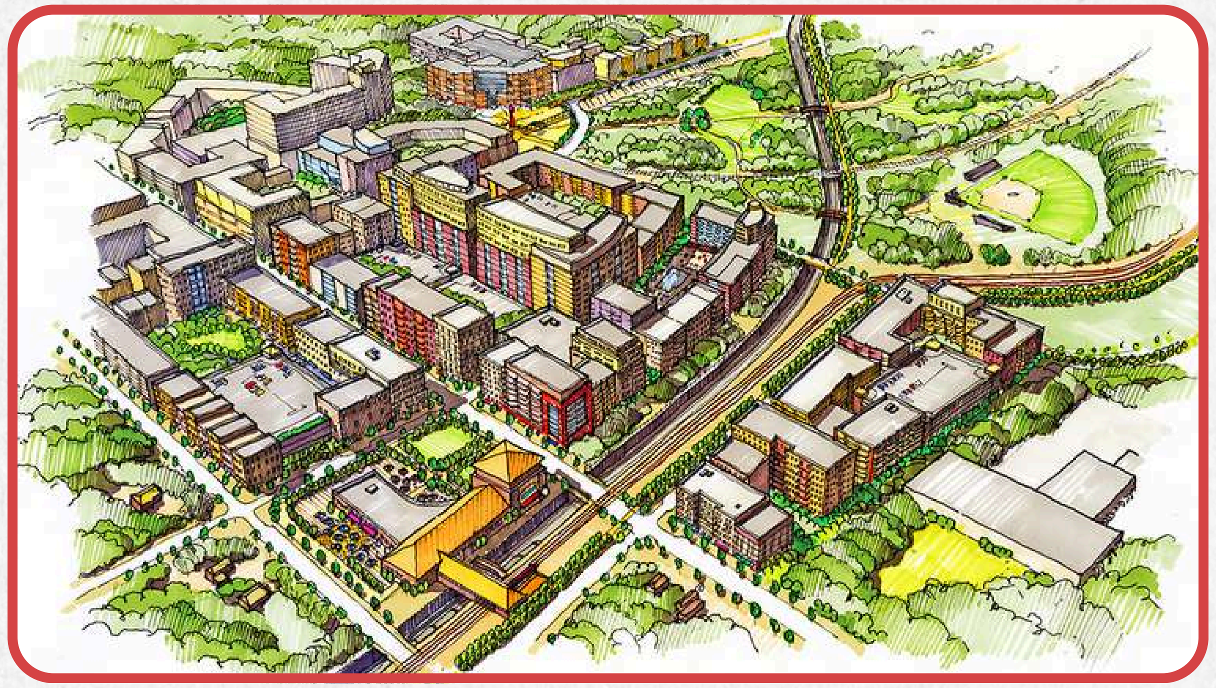
**MY WEBSITE: CLICK HERE FOR MY PORTOFOLIO**

# EDUCATION

## URBAN PLANNING

Urban and Regional Planning is a multidisciplinary field that focuses on designing and managing sustainable cities and regions. The program equips students with knowledge in spatial planning, land use, transportation, and urban design, supported by geospatial technology and data analysis.

PERENCANAAN WILAYAH DAN KOTA



- Urban Planning & Design Laboratory – Focuses on urban form, spatial planning, and urban design strategies.
- Regional Planning Laboratory – Explores regional development, policy, and spatial analysis at a wider territorial scale.
- Computation & Modeling Laboratory – Specializes in GIS, remote sensing, urban modeling, and transportation planning.

# MY FOCUS

## URBAN PLANNING STUDIO

Engaged in collaborative projects focusing on spatial planning, land use strategies, and urban design to create sustainable and livable cities.

LAPORAN

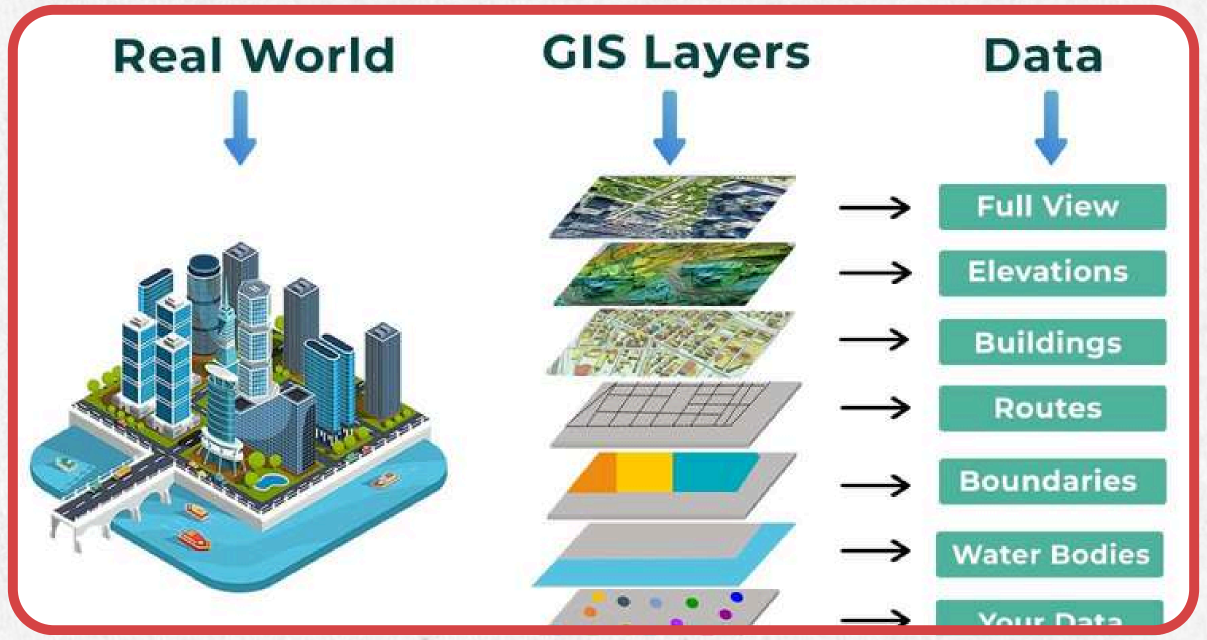
PENDAHULUAN

RZWP3K

*"Pengembangan Kawasan Kuliner Berbasis Perikanan Tangkap Wilayah Perencanaan Kecamatan Palang dan Kecamatan Tuban Kabupaten Tuban"*

## GIS AND TRASNPORTATION ANALYST

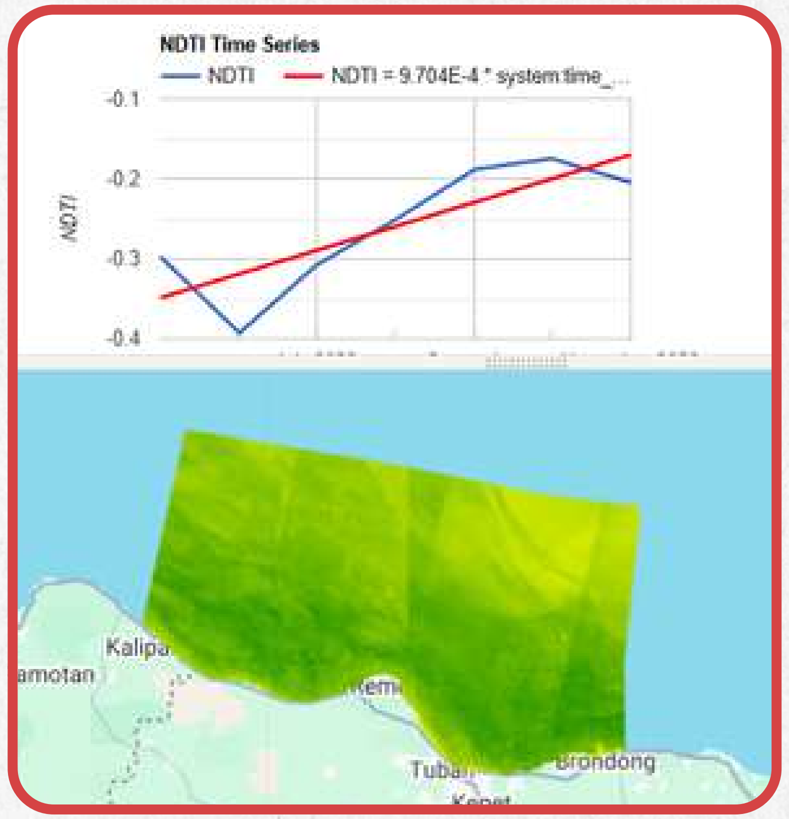
Specialized in geospatial analysis, land use modeling, and transportation planning using tools such as ArcGIS, Google Earth Engine, and GeoSOS-FLUS.


Real World
GIS Layers
Data
Full View
Elevations
Buildings
Routes
Boundaries
Water Bodies

# SKILL

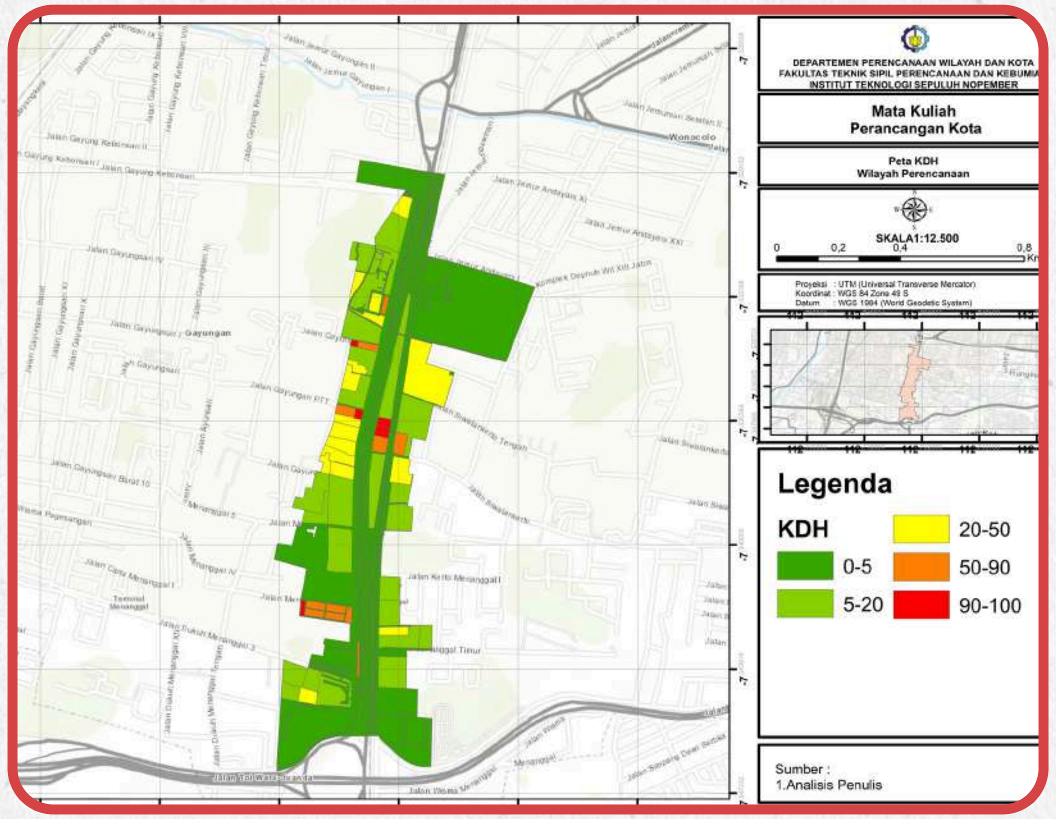
## REMOTE SENSING

Conducting coastal and land cover analysis using Google Earth Engine (GEE) with indices such as NDTI, NDBI, NDWI, and others, as well as processing LiDAR data for spatial modeling and environmental monitoring.



## GIS

Performing spatial data management and transportation analysis, including accessibility studies, land use modeling, and geospatial visualization using ArcGIS and other GIS tools.


DEPARTEMEN PERENCANAAN WILAYAH DAN KOTA
FAKULTAS TEKNIK SIPIL PERENCANAAN DAN KEBUMIA
INSTITUT TEKNOLOGI SEPULUH NOPEMBER
Mata Kuliah
Perancangan Kota
Peta KDH
Wilayah Perencanaan
SKALA1:12.500
0
0,2
0,4
0,8
Proyeksi : UTM (Universal Transverse Mercator)
Koordinat : WGS 84 Zona 49 S
Datum : WGS 1984 (World Geodetic System)
Legenda
KDH
🟩 0-5
🟩 5-20
🟨 20-50
🟧 50-90
🟥 90-100
Sumber :
1.Analisis Penulis

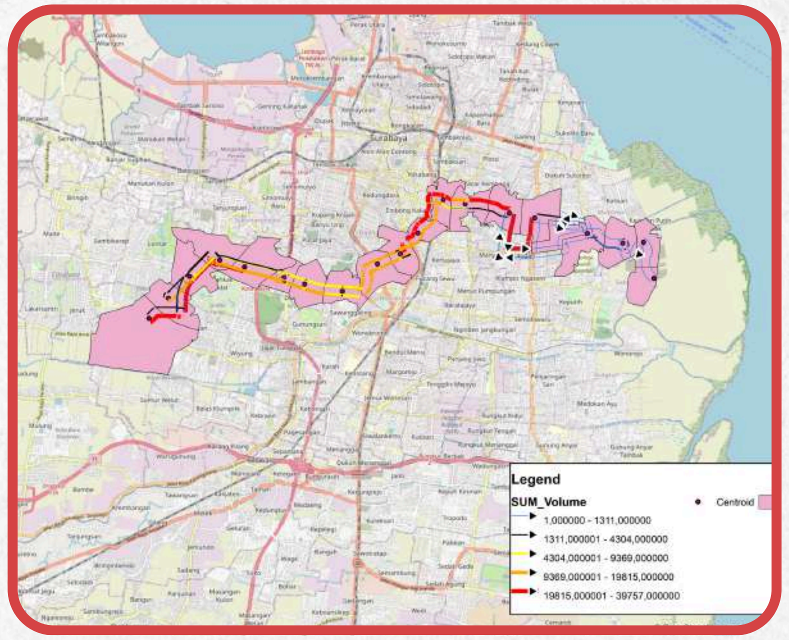
## URBAN DESIGN

Designing urban spaces and public areas with a focus on livability, walkability, and sustainability, integrating planning concepts into visual and practical design outputs.

# WORK EXPERIENCE

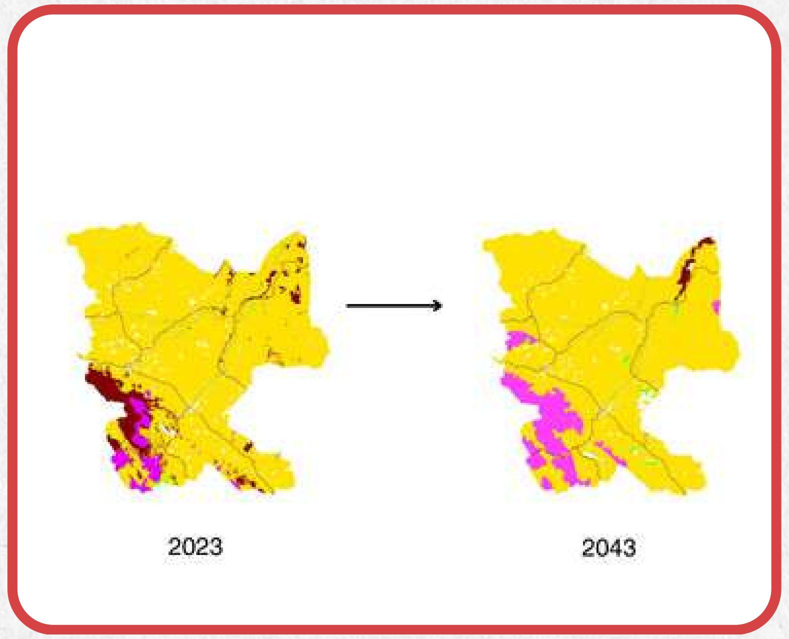
## MRT SURVEYOR

Conducted household surveys and traffic counting to support urban mobility and transportation planning studies, including calculations using the Four Step Model (trip generation, trip distribution, mode choice, and route assignment).


Legend
SUM_Volume
1,000000 - 1311,000000
1311,000001 - 4304,000000
4304,000001 - 9369,000000
9369,000001 - 19815,000000
19815,000001 - 39757,000000
Centroid

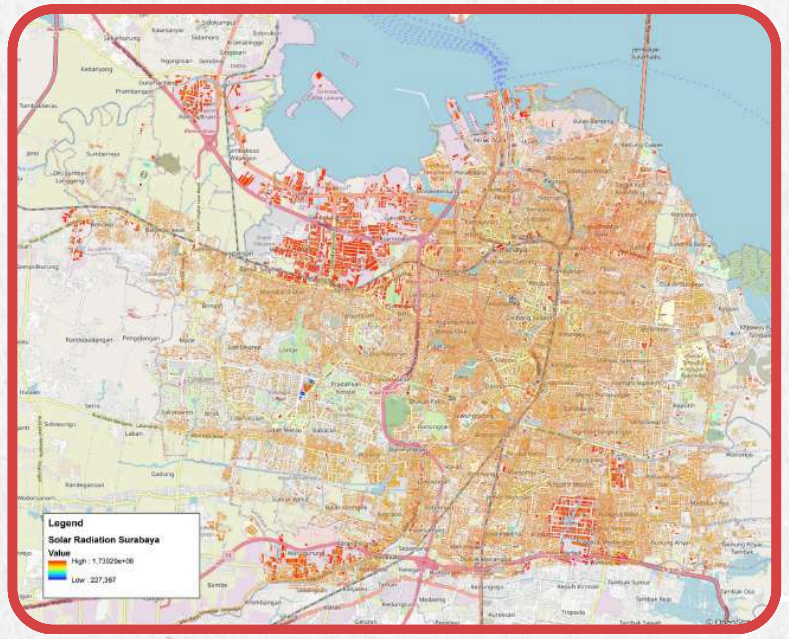
## RESEARCH ANALYSIS

Performed land use change modeling using GeoSOS-FLUS in Rokan Hulu, focusing on spatial dynamics and future urban growth projection.

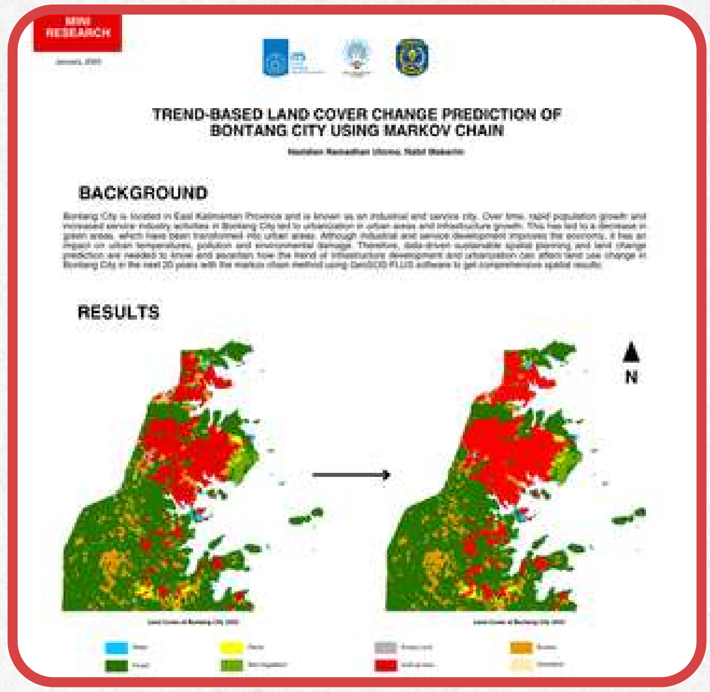


## GREENOVA

Assisted in spatial analysis for solar energy potential modeling in Surabaya by processing LiDAR data into nDSM to identify suitable areas for solar panel installation.


Legend
Solar Radiation Surabaya
Value
High : 1,73929e+06
Low : 227,367

# PROJECT PORTFOLIO

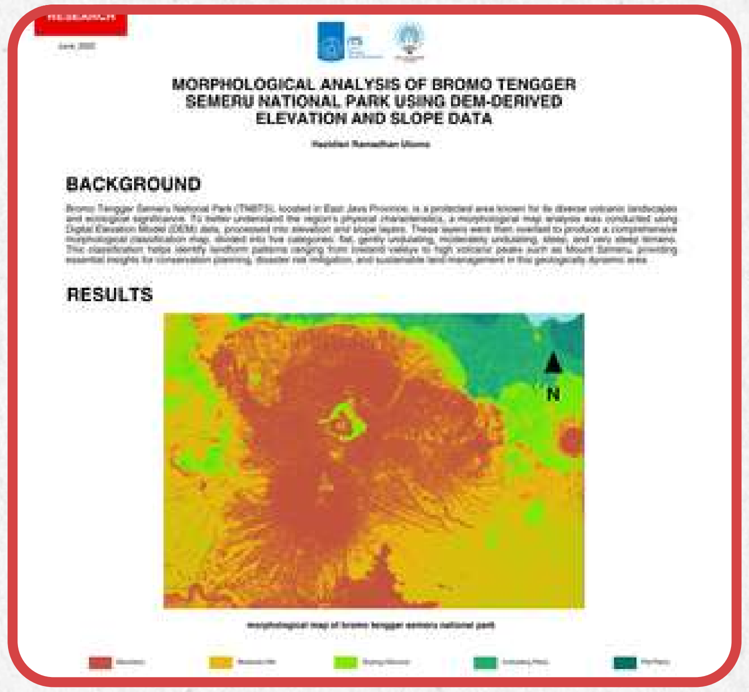


MINI RESEARCH

### TREND-BASED LAND COVER CHANGE PREDICTION OF BONTANG CITY USING MARKOV CHAIN

#### BACKGROUND

#### RESULTS

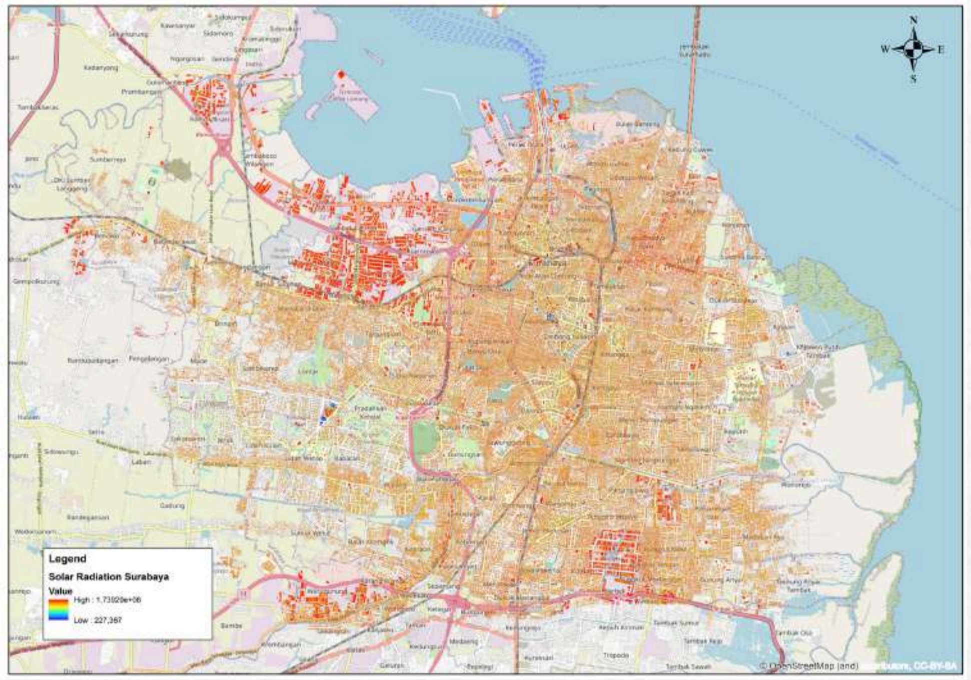


RESEARCH

### MORPHOLOGICAL ANALYSIS OF BROMO TENGGER SEMERU NATIONAL PARK USING DEM-DERIVED ELEVATION AND SLOPE DATA

#### BACKGROUND

#### RESULTS



Legend

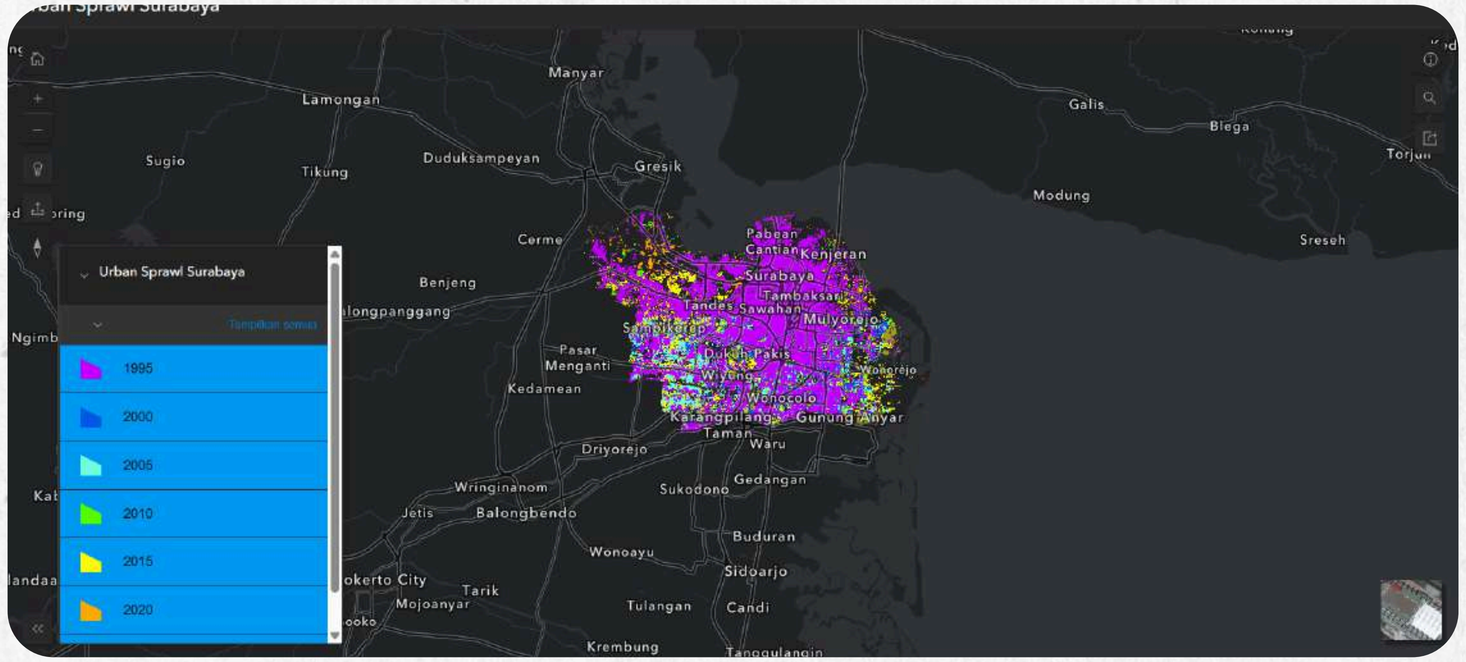
Solar Radiation Surabaya

Value

High : 1,73329e+06

Low : 227,367

© OpenStreetMap (and) contributors, CC-BY-SA

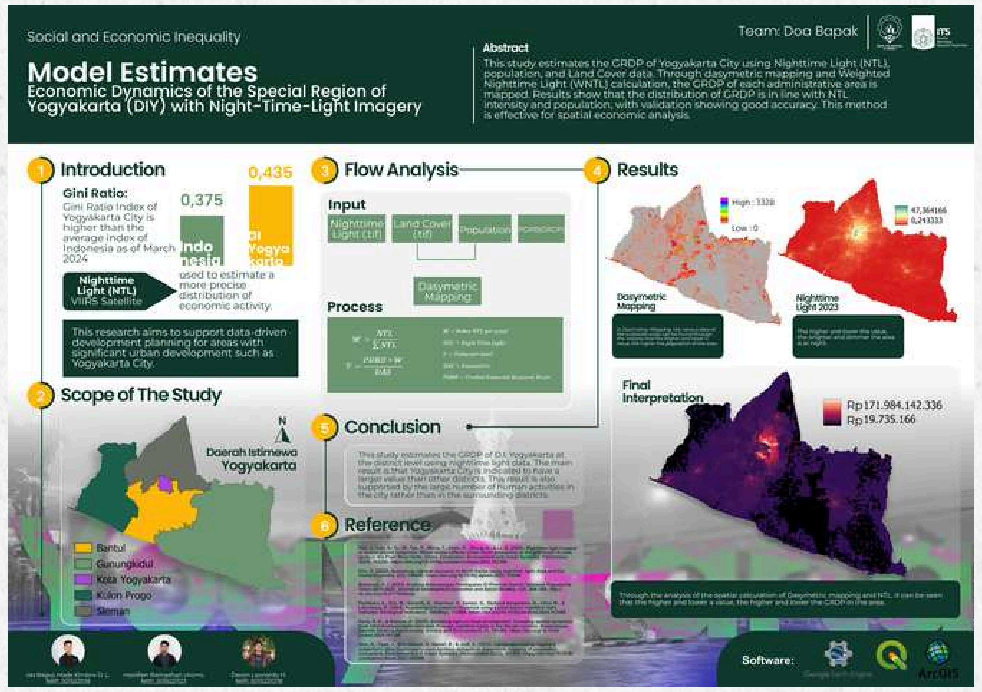


Urban Sprawl Surabaya

Tampilkan semua

- 1995
- 2000
- 2005
- 2010
- 2015
- 2020

Lamongan
Manyar
Sugio
Tikung
Duduksampeyan
Gresik
Galis
Blega
Modung
Sreseh
Cerme
Benjeng
Kenjeran
Surabaya
Tambaksari
Mulyorejo
Pasar
Menganti
Kedamean
Driyorejo
Taman
Waru
Gedangan
Sukodono
Wringinanom
Balongbendo
Wonoayu
Buduran
Sidoarjo
Tulangan
Candi
Krembung
Tarik
Mojoanyar
Jetis



Social and Economic Inequality

## Model Estimates

### Economic Dynamics of the Special Region of Yogyakarta (DIY) with Night-Time-Light Imagery

Team: Doa Bapak

**Abstract**

This study estimates the GRDP of Yogyakarta City using Nighttime Light (NTL), population, and Land Cover data. Through dasymetric mapping and Weighted Nighttime Light (WNTL) calculation, the GRDP of each administrative area is mapped. Results show that the distribution of GRDP is in line with NTL intensity and population, with validation showing good accuracy. This method is effective for spatial economic analysis.

#### 1 Introduction

**Gini Ratio:**
Gini Ratio Index of Yogyakarta City is higher than the average index of Indonesia as of March 2024

0,375
Indonesia
0,435
DI Yogyakarta

Nighttime Light (NTL) VIIRS Satellite used to estimate a more precise distribution of economic activity.

This research aims to support data-driven development planning for areas with significant urban development such as Yogyakarta City.

#### 2 Scope of The Study

N
Daerah Istimewa Yogyakarta
Bantul
Gunungkidul
Kota Yogyakarta
Kulon Progo
Sleman

#### 3 Flow Analysis

Input
- Nighttime Light (tif)
- Land Cover (tif)
- Population
- PDRB/GRDP

Dasymetric Mapping

Process

#### 4 Results

High : 3328
Low : 0
Dasymetric Mapping
47,364166
0,243333
Nighttime Light 2023
Final Interpretation
Rp 171.984.142.336
Rp 19.735.166

#### 5 Conclusion

This study estimates the GRDP of D.I. Yogyakarta at the district level using nighttime light data. The main result is that Yogyakarta City is indicated to have a larger value than other districts. This result is also supported by the large number of human activities in the city rather than in the surrounding districts.

#### 6 Reference

Software:

# PROJECT 1

## LAND COVER CHANGES AT BONTANG CITY

This project predicts land cover change in Bontang City over the next 20 years using the Markov Chain method in GeoSOS-FLUS. Results show an 8.28 km² increase in built-up areas and significant loss of forests and vegetation due to urbanization and industrial growth. Recommendations include promoting vertical housing, preserving green areas, and improving waste management

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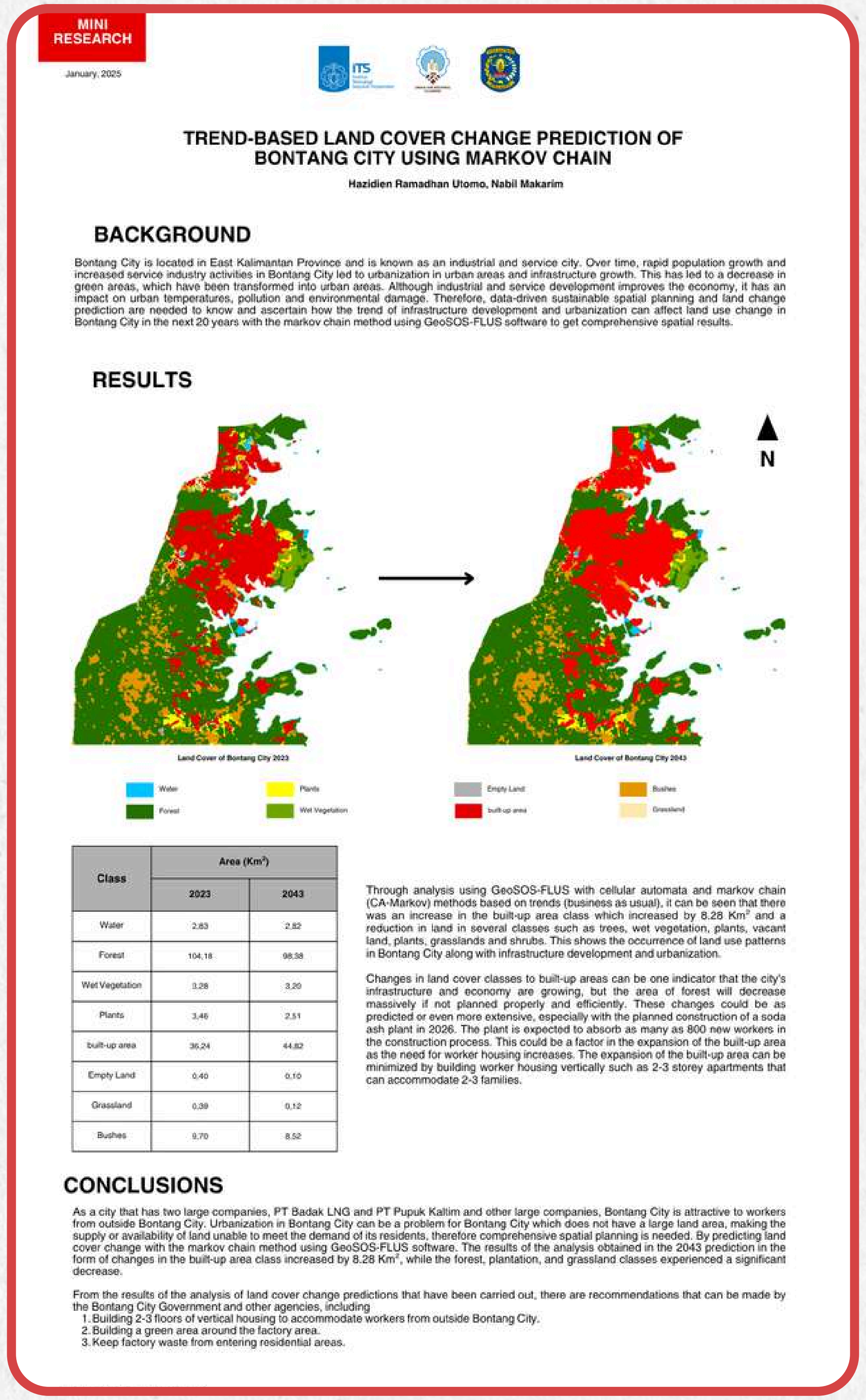

**MINI RESEARCH**

January, 2025

### TREND-BASED LAND COVER CHANGE PREDICTION OF BONTANG CITY USING MARKOV CHAIN

**Hazidien Ramadhan Utomo, Nabil Makarim**

#### BACKGROUND

Bontang City is located in East Kalimantan Province and is known as an industrial and service city. Over time, rapid population growth and increased service industry activities in Bontang City led to urbanization in urban areas and infrastructure growth. This has led to a decrease in green areas, which have been transformed into urban areas. Although industrial and service development improves the economy, it has an impact on urban temperatures, pollution and environmental damage. Therefore, data-driven sustainable spatial planning and land change prediction are needed to know and ascertain how the trend of infrastructure development and urbanization can affect land use change in Bontang City in the next 20 years with the markov chain method using GeoSOS-FLUS software to get comprehensive spatial results.

#### RESULTS

| Class | Area (Km²) | |
|---|---|---|
| | 2023 | 2043 |
| Water | 2,83 | 2,82 |
| Forest | 104,18 | 98,38 |
| Wet Vegetation | 3,28 | 3,20 |
| Plants | 3,46 | 2,51 |
| built-up area | 36,24 | 44,82 |
| Empty Land | 0,40 | 0,10 |
| Grassland | 0,39 | 0,12 |
| Bushes | 9,70 | 8,52 |

Through analysis using GeoSOS-FLUS with cellular automata and markov chain (CA-Markov) methods based on trends (business as usual), it can be seen that there was an increase in the built-up area class which increased by 8.28 Km² and a reduction in land in several classes such as trees, wet vegetation, plants, vacant land, plants, grasslands and shrubs. This shows the occurrence of land use patterns in Bontang City along with infrastructure development and urbanization.

Changes in land cover classes to built-up areas can be one indicator that the city's infrastructure and economy are growing, but the area of forest will decrease massively if not planned properly and efficiently. These changes could be as predicted or even more extensive, especially with the planned construction of a soda ash plant in 2026. The plant is expected to absorb as many as 800 new workers in the construction process. This could be a factor in the expansion of the built-up area as the need for worker housing increases. The expansion of the built-up area can be minimized by building worker housing vertically such as 2-3 storey apartments that can accommodate 2-3 families.

#### CONCLUSIONS

As a city that has two large companies, PT Badak LNG and PT Pupuk Kaltim and other large companies, Bontang City is attractive to workers from outside Bontang City. Urbanization in Bontang City can be a problem for Bontang City which does not have a large land area, making the supply or availability of land unable to meet the demand of its residents, therefore comprehensive spatial planning is needed. By predicting land cover change with the markov chain method using GeoSOS-FLUS software. The results of the analysis obtained in the 2043 prediction in the form of changes in the built-up area class increased by 8.28 Km², while the forest, plantation, and grassland classes experienced a significant decrease.

From the results of the analysis of land cover change predictions that have been carried out, there are recommendations that can be made by the Bontang City Government and other agencies, including

1. Building 2-3 floors of vertical housing to accommodate workers from outside Bontang City.
2. Building a green area around the factory area.
3. Keep factory waste from entering residential areas.

# PROJECT 2

## MORPHOLOGICAL ANALYSIS OF BROMO TENGGER SEMERU NATIONAL PARK USING DEM-DERIVED ELEVATION AND SLOPE DATA

This study maps the morphology of Bromo Tengger Semeru National Park using DEM-derived elevation and slope data. Results show that over 75% of the area is dominated by steep and moderately steep terrain, shaped by volcanic activity around Mount Semeru and Bromo. These findings provide essential insights for conservation planning, disaster risk mitigation, and sustainable land management.

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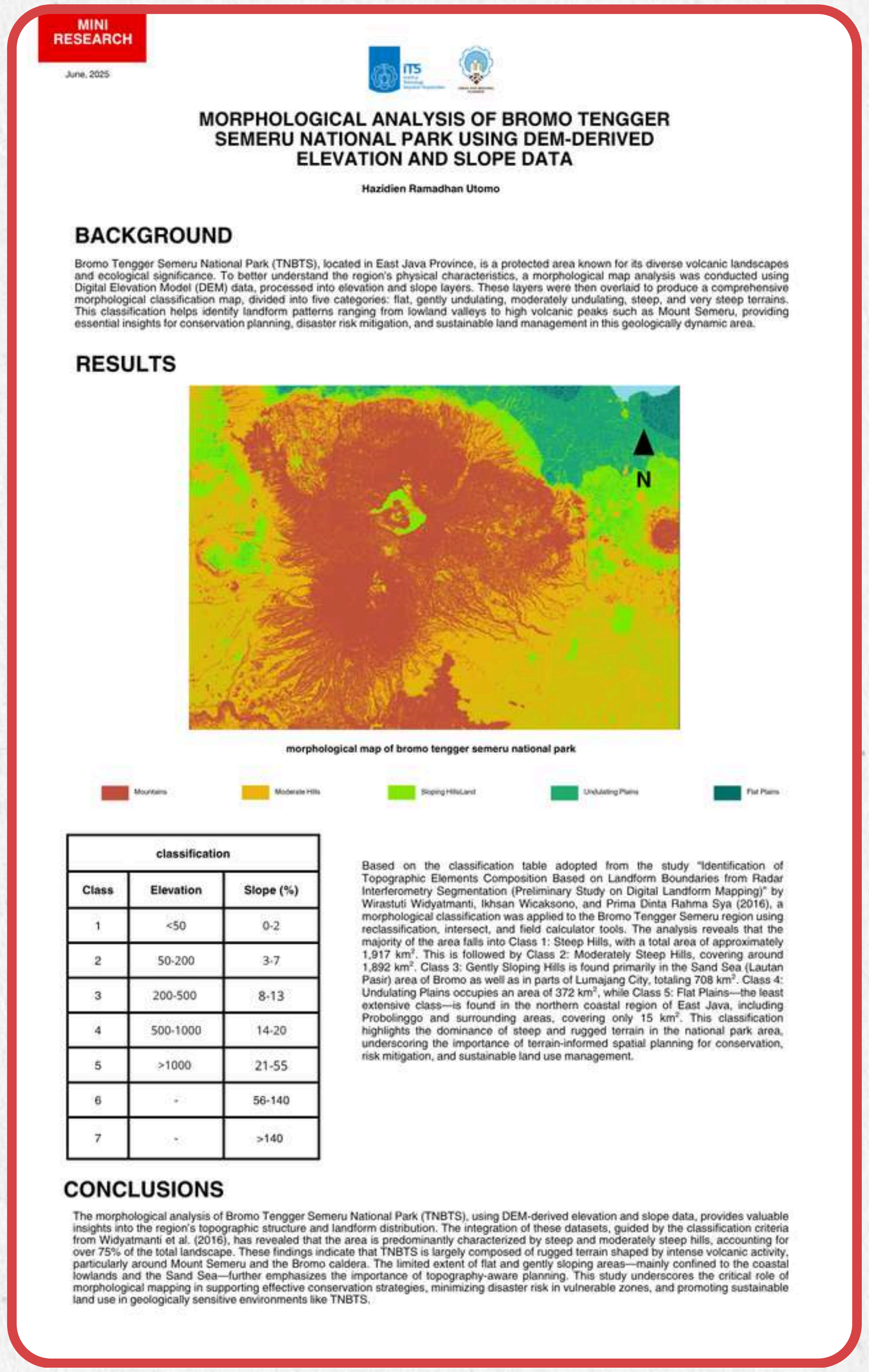


MINI RESEARCH

June, 2025

### MORPHOLOGICAL ANALYSIS OF BROMO TENGGER SEMERU NATIONAL PARK USING DEM-DERIVED ELEVATION AND SLOPE DATA

**Hazidien Ramadhan Utomo**

#### BACKGROUND

Bromo Tengger Semeru National Park (TNBTS), located in East Java Province, is a protected area known for its diverse volcanic landscapes and ecological significance. To better understand the region's physical characteristics, a morphological map analysis was conducted using Digital Elevation Model (DEM) data, processed into elevation and slope layers. These layers were then overlaid to produce a comprehensive morphological classification map, divided into five categories: flat, gently undulating, moderately undulating, steep, and very steep terrains. This classification helps identify landform patterns ranging from lowland valleys to high volcanic peaks such as Mount Semeru, providing essential insights for conservation planning, disaster risk mitigation, and sustainable land management in this geologically dynamic area.

#### RESULTS

morphological map of bromo tengger semeru national park

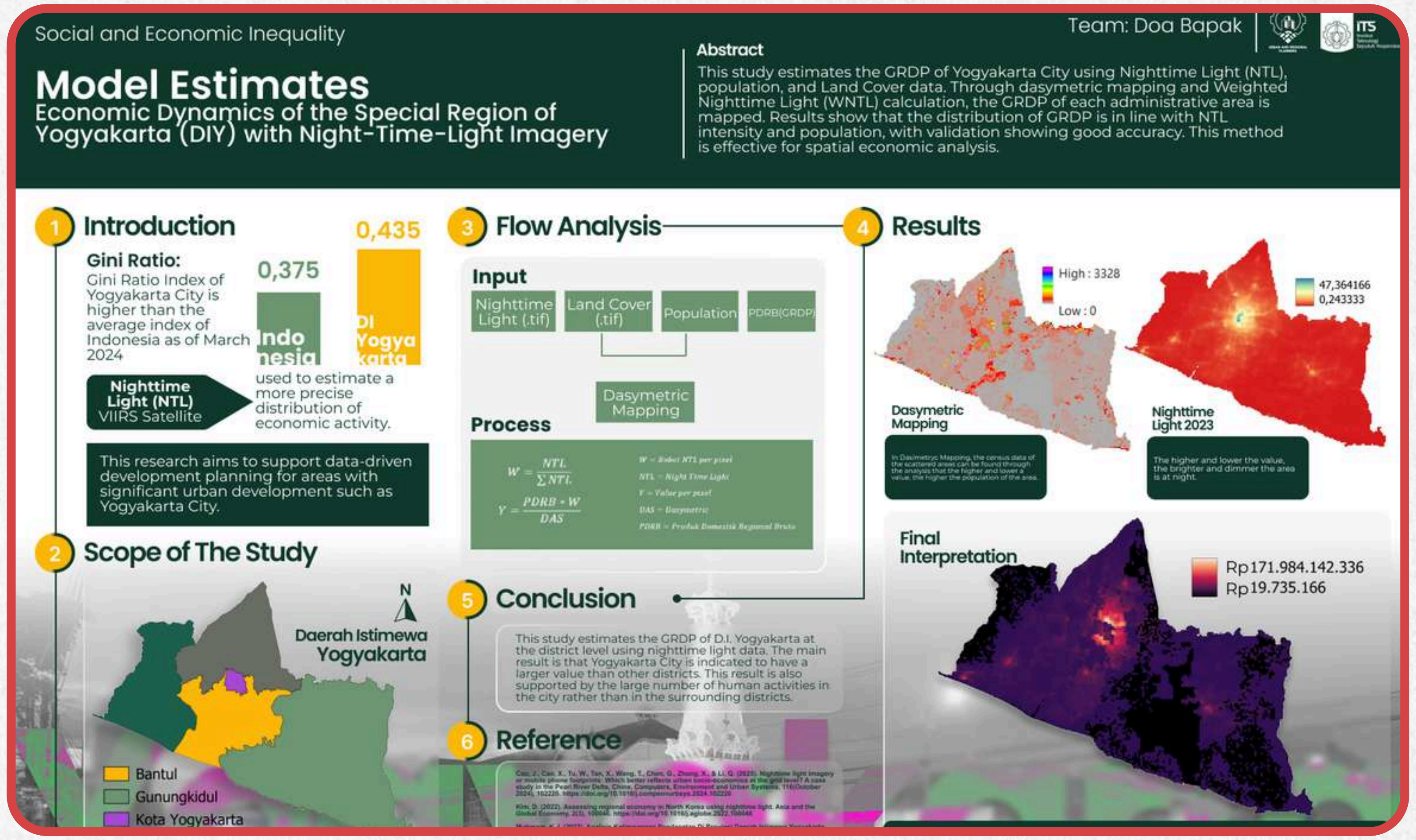
| classification | | |
|---|---|---|
| Class | Elevation | Slope (%) |
| 1 | <50 | 0-2 |
| 2 | 50-200 | 3-7 |
| 3 | 200-500 | 8-13 |
| 4 | 500-1000 | 14-20 |
| 5 | >1000 | 21-55 |
| 6 | - | 56-140 |
| 7 | - | >140 |

Based on the classification table adopted from the study "Identification of Topographic Elements Composition Based on Landform Boundaries from Radar Interferometry Segmentation (Preliminary Study on Digital Landform Mapping)" by Wirastuti Widyatmanti, Ikhsan Wicaksono, and Prima Dinta Rahma Sya (2016), a morphological classification was applied to the Bromo Tengger Semeru region using reclassification, intersect, and field calculator tools. The analysis reveals that the majority of the area falls into Class 1: Steep Hills, with a total area of approximately 1,917 km². This is followed by Class 2: Moderately Steep Hills, covering around 1,892 km². Class 3: Gently Sloping Hills is found primarily in the Sand Sea (Lautan Pasir) area of Bromo as well as in parts of Lumajang City, totaling 708 km². Class 4: Undulating Plains occupies an area of 372 km², while Class 5: Flat Plains—the least extensive class—is found in the northern coastal region of East Java, including Probolinggo and surrounding areas, covering only 15 km². This classification highlights the dominance of steep and rugged terrain in the national park area, underscoring the importance of terrain-informed spatial planning for conservation, risk mitigation, and sustainable land use management.

#### CONCLUSIONS

The morphological analysis of Bromo Tengger Semeru National Park (TNBTS), using DEM-derived elevation and slope data, provides valuable insights into the region's topographic structure and landform distribution. The integration of these datasets, guided by the classification criteria from Widyatmanti et al. (2016), has revealed that the area is predominantly characterized by steep and moderately steep hills, accounting for over 75% of the total landscape. These findings indicate that TNBTS is largely composed of rugged terrain shaped by intense volcanic activity, particularly around Mount Semeru and the Bromo caldera. The limited extent of flat and gently sloping areas—mainly confined to the coastal lowlands and the Sand Sea—further emphasizes the importance of topography-aware planning. This study underscores the critical role of morphological mapping in supporting effective conservation strategies, minimizing disaster risk in vulnerable zones, and promoting sustainable land use in geologically sensitive environments like TNBTS.

# PROJECT 3



Social and Economic Inequality

Team: Doa Bapak

**Model Estimates**
Economic Dynamics of the Special Region of Yogyakarta (DIY) with Night-Time-Light Imagery

**Abstract**
This study estimates the GRDP of Yogyakarta City using Nighttime Light (NTL), population, and Land Cover data. Through dasymetric mapping and Weighted Nighttime Light (WNTL) calculation, the GRDP of each administrative area is mapped. Results show that the distribution of GRDP is in line with NTL intensity and population, with validation showing good accuracy. This method is effective for spatial economic analysis.

**1 Introduction**

**Gini Ratio:** Gini Ratio Index of Yogyakarta City is higher than the average index of Indonesia as of March 2024

**Nighttime Light (NTL)** VIIRS Satellite used to estimate a more precise distribution of economic activity.

This research aims to support data-driven development planning for areas with significant urban development such as Yogyakarta City.

**2 Scope of The Study**

**3 Flow Analysis**

**Input**

**Process**

$$W = \frac{NTL}{\sum NTL}$$

$$Y = \frac{PDRB \cdot W}{DAS}$$

W = Bobot NTL per pixel
NTL = Night Time Light
Y = Value per pixel
DAS = Dasymetric
PDRB = Produk Domestik Regional Bruto

**4 Results**

**Dasymetric Mapping**

In Dasymetric Mapping, the census data of the scattered areas can be found through the analysis that the higher and lower a value, the higher the population of the area.

**Nighttime Light 2023**

The higher and lower the value, the brighter and dimmer the area is at night.

**Final Interpretation**

**5 Conclusion**

This study estimates the GRDP of D.I. Yogyakarta at the district level using nighttime light data. The main result is that Yogyakarta City is indicated to have a larger value than other districts. This result is also supported by the large number of human activities in the city rather than in the surrounding districts.

**6 Reference**

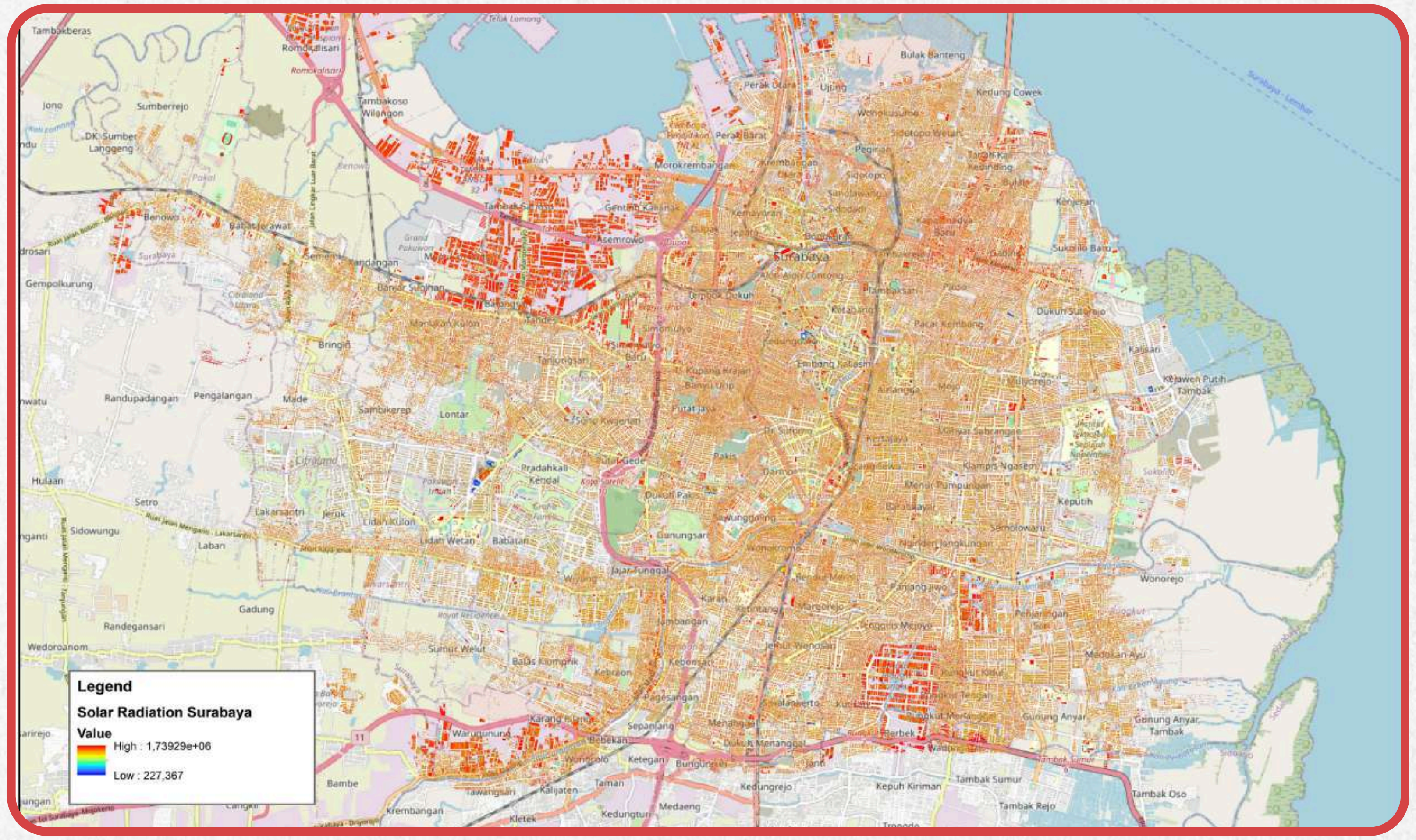
## MODEL ESTIMATES
## ECONOMIC DYNAMICS OF THE SPECIAL REGION OF YOGYAKARTA (DIY) WITH NIGHT-TIME-LIGHT IMAGERY

This study estimates the GRDP of Yogyakarta Special Region using Night-Time Light (NTL) imagery, land cover, and population data with dasymetric mapping. Results show that Yogyakarta City has the highest economic activity compared to other districts, supported by its high urban intensity. The findings demonstrate that NTL is effective for spatial economic analysis and supports data-driven development planning.

***CLICK HERE TO ACCESS THE FULL POSTER***

# PROJECT 4

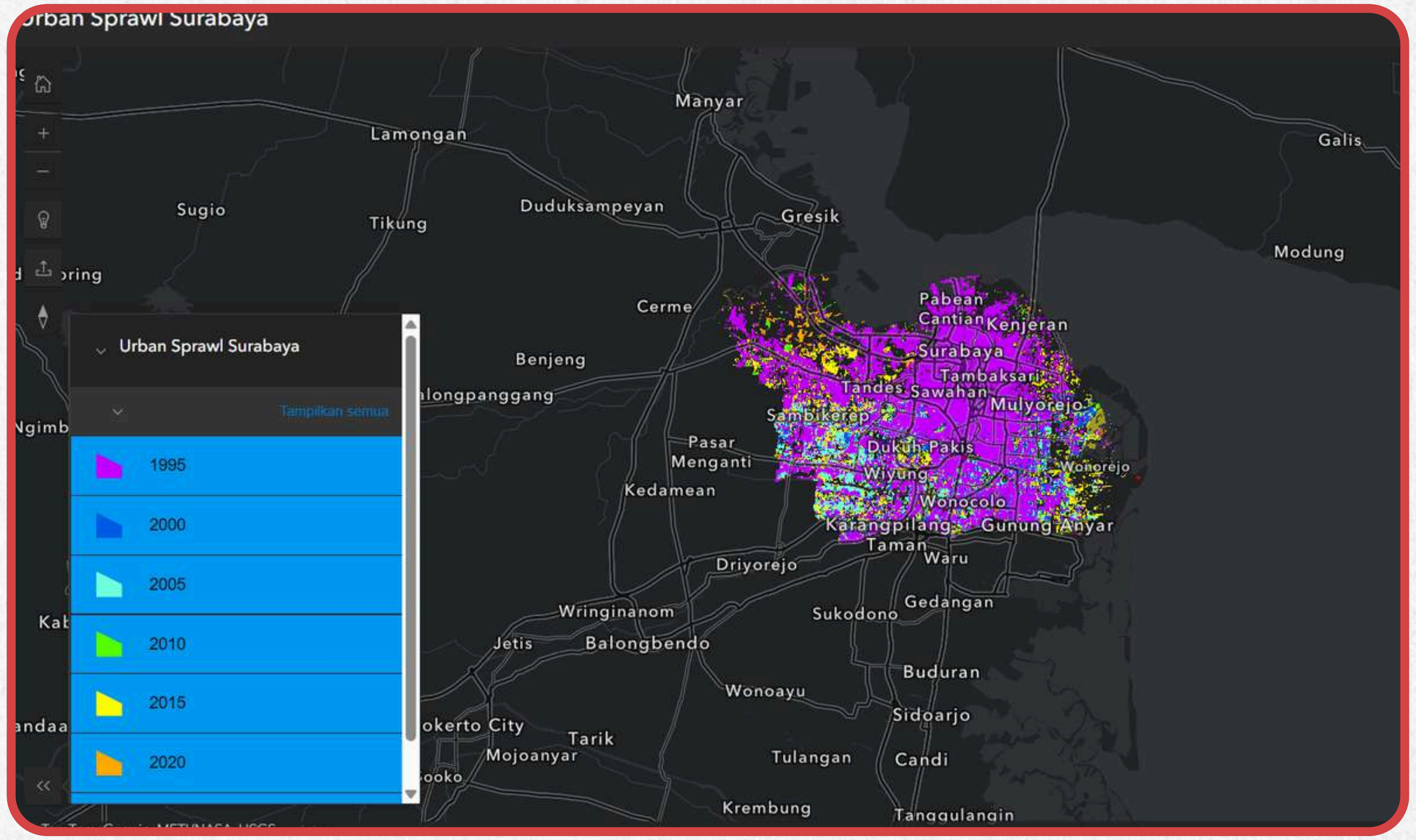
## SOLAR ENERGY POTENTIAL ANALYSIS IN SURABAYA USING LIDAR-DERIVED NDSM AND ARCGIS SOLAR RADIATION TOOL



This project analyzes the solar energy potential of Surabaya using LiDAR-derived nDSM processed with the ArcGIS Area Solar Radiation tool, mapping annual rooftop solar potential and assessing spatial variability of solar irradiance (kWh/m²/year) by considering building orientation, shading, land cover, and urban density, and further estimating rooftop photovoltaic electricity generation potential of up to approximately 234 kWh/m²/year per panel based on solar radiation data, module efficiency, and performance ratio assumptions.

# PROJECT 5

creative portfolio

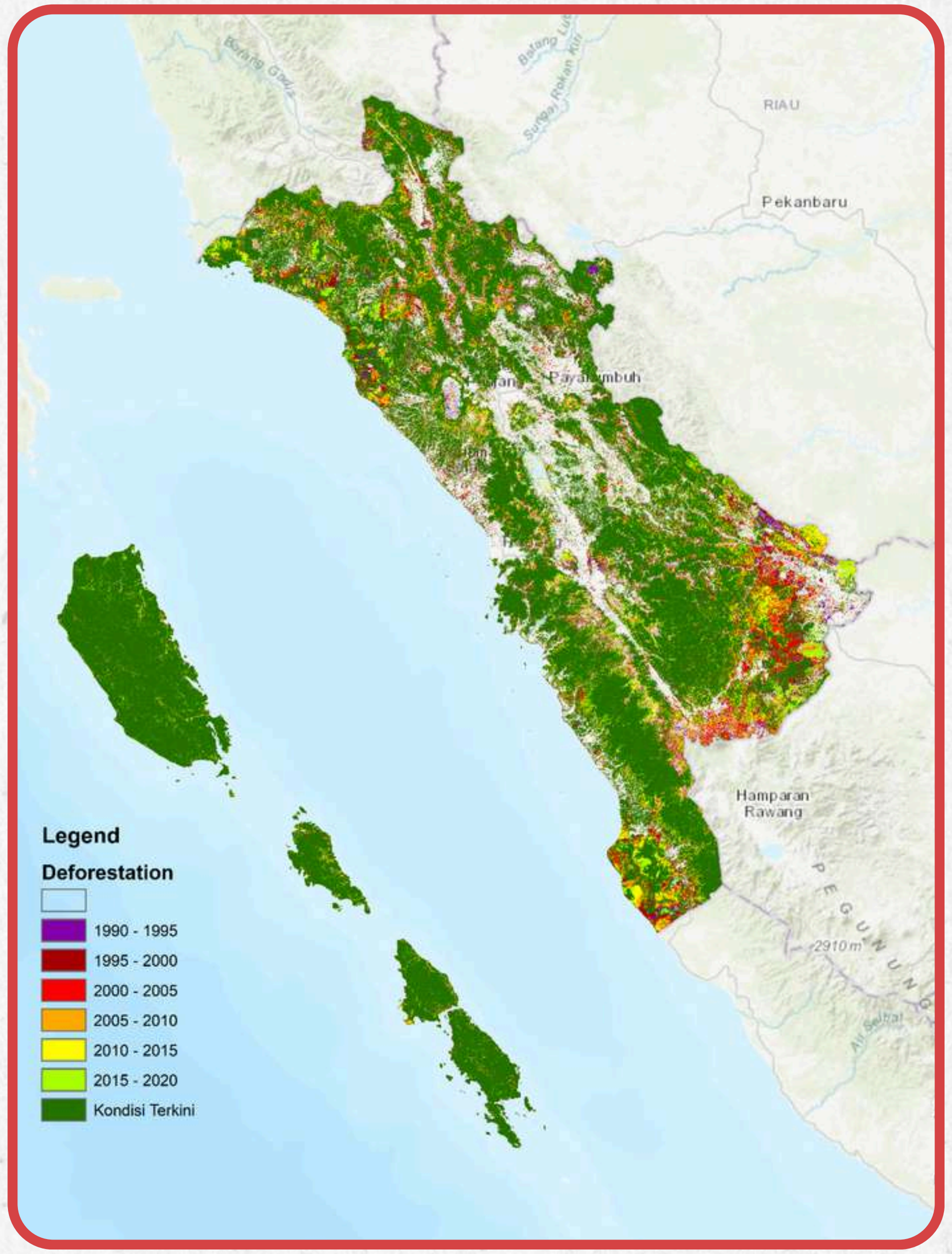

Urban Sprawl Surabaya
Manyar
Lamongan
Galis
Sugio
Tikung
Duduksampeyan
Gresik
Modung
Cerme
Pabean Cantian
Kenjeran
Urban Sprawl Surabaya
Tampilkan semua
Benjeng
Surabaya
Tambaksari
Tandes
Sawahan
Mulyorejo
Sambikerep
1995
2000
2005
2010
2015
2020
Pasar Menganti
Dukuh Pakis
Kedamean
Wiyung
Wonorejo
Wonocolo
Karangpilang
Gunung Anyar
Taman
Waru
Driyorejo
Gedangan
Sukodono
Wringinanom
Jetis
Balongbendo
Buduran
Wonoayu
Sidoarjo
Tarik
Mojoanyar
Tulangan
Candi
Krembung
Tanggulangin

## URBAN EXPANSION MAP AT SURABAYA CITY

This map shows the growth and distribution of population in Surabaya City from 1995 to 2025. The distribution of the population tends to point to the east and north. From this map, it is expected that stakeholders can make more appropriate policies in the future.

***CLICK HERE TO ACCESS INTERACTIVE MAP***

# PROJECT 6

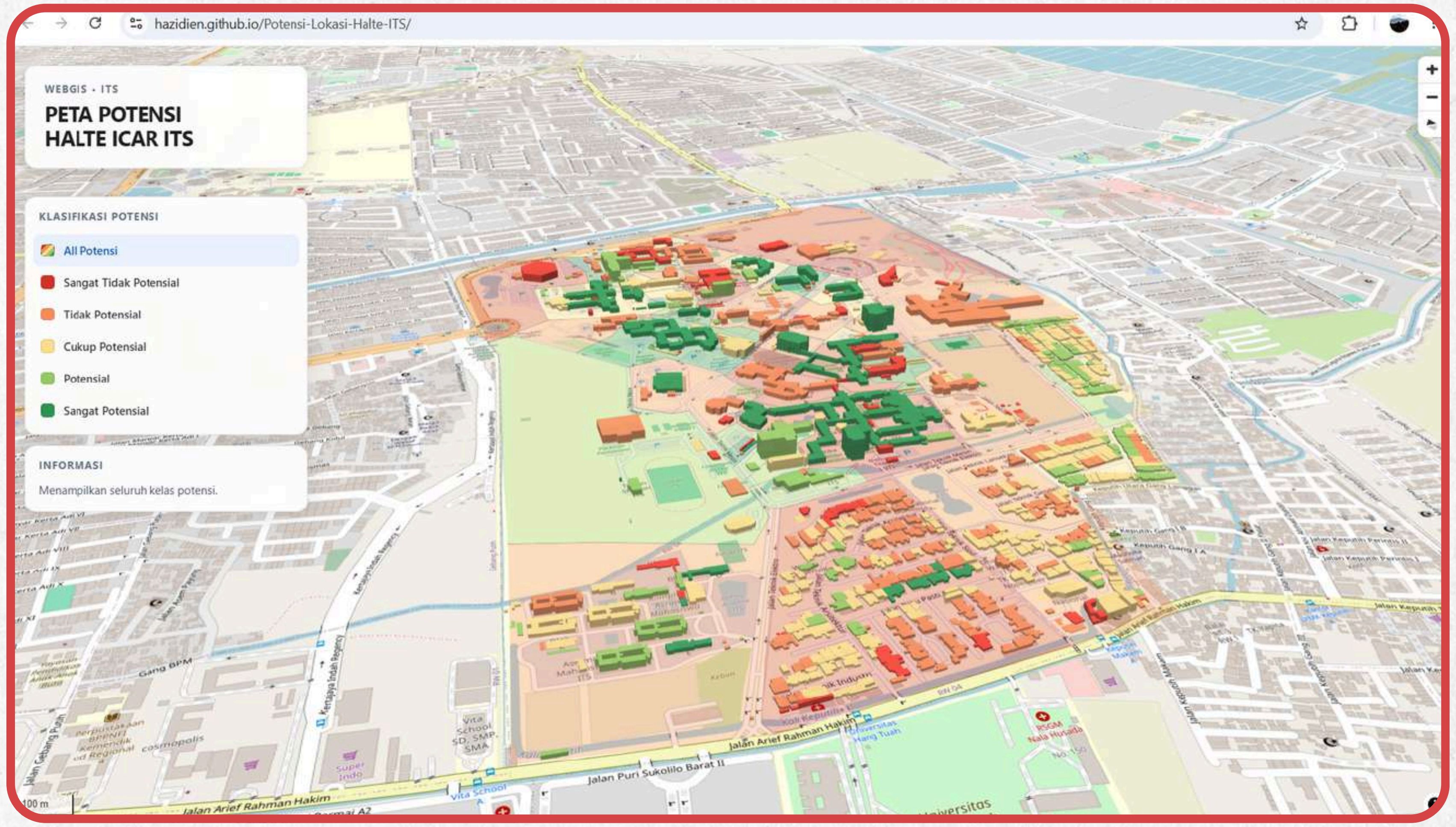

RIAU
Pekanbaru
Payakumbuh
Hamparan Rawang
2910 m
Legend
Deforestation
1990 - 1995
1995 - 2000
2000 - 2005
2005 - 2010
2010 - 2015
2015 - 2020
Kondisi Terkini

## SPATIO-TEMPORAL DEFORESTATION MODELING PROJECT – WEST SUMATRA

The Spatio-Temporal Deforestation Modeling Project in West Sumatra (Nov 2025–Dec 2025) involved analyzing multitemporal Landsat imagery from 2009 to 2024 using Google Earth Engine to identify patterns of forest loss, degradation, and regrowth, applying vegetation index–based thresholding and the Normalized Difference Fraction Index (NDFI) to detect deforestation dynamics, hotspots, and temporal change trajectories, and producing spatial outputs such as deforestation maps, hotspot analyses, and comparative method evaluations to support environmental monitoring and spatial planning decision-making.

***CLICK HERE TO ACCESS PROJECT REPORT***

# PROJECT 7


hazidien.github.io/Potensi-Lokasi-Halte-ITS/
WEBGIS • ITS
PETA POTENSI
HALTE ICAR ITS
KLASIFIKASI POTENSI
All Potensi
🟥 Sangat Tidak Potensial
🟧 Tidak Potensial
🟨 Cukup Potensial
🟩 Potensial
🟩 Sangat Potensial
INFORMASI
100 m

Menampilkan seluruh kelas potensi.

## SOLAR ENERGY POTENTIAL ANALYSIS IN SURABAYA USING LIDAR-DERIVED NDSM AND ARCGIS SOLAR RADIATION TOOL

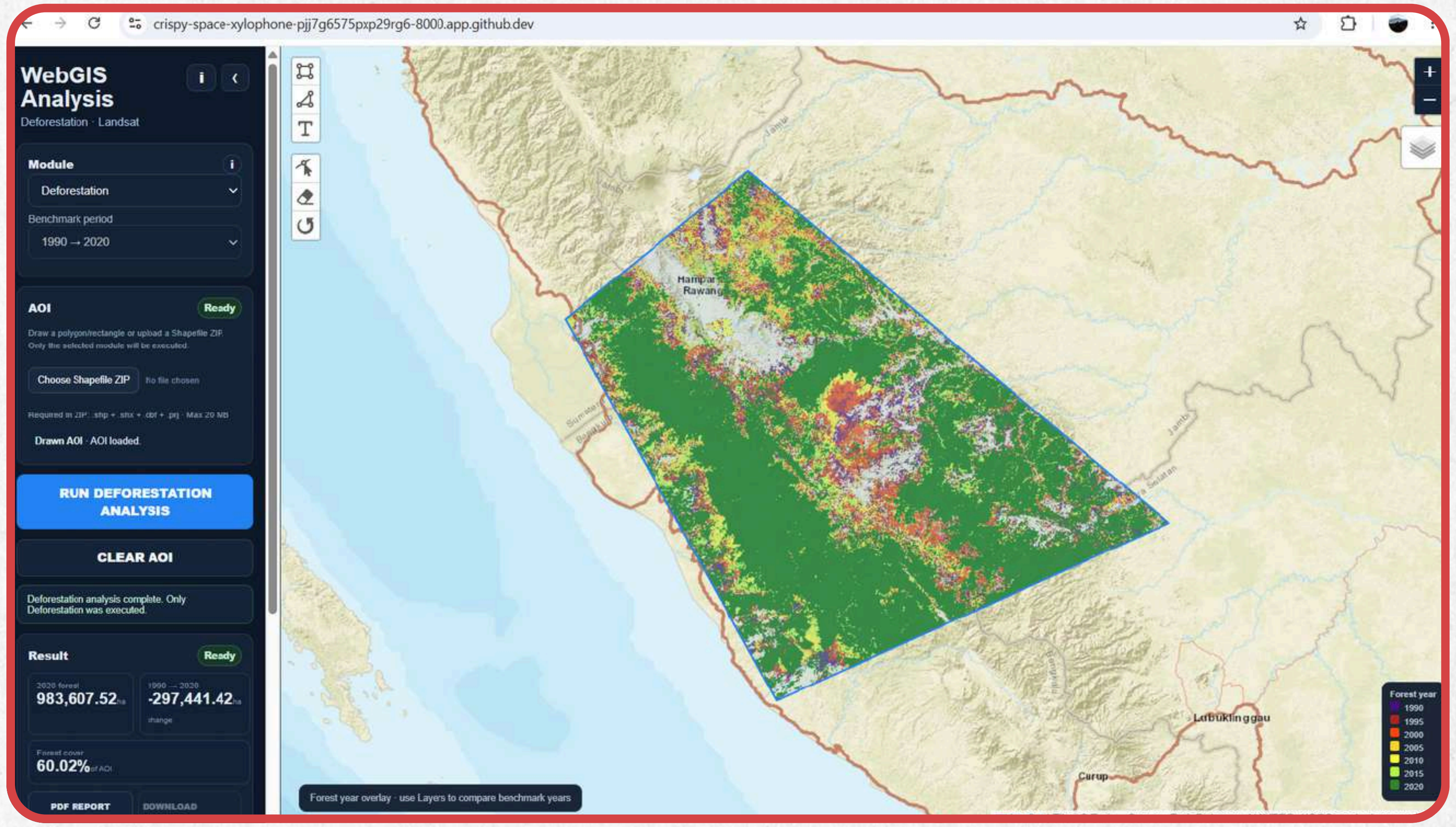
Developed a 3D WebGIS using Urban Network Analysis (UNA) and Service Network Analysis (SNA) to identify buildings with high potential for campus transit stops across ITS, PENS, and PPNS. The analysis evaluates network accessibility and service coverage to support integrated campus transportation planning, with results visualized interactively through a 3D MapLibre WebGIS.

**click to acces**

# PROJECT 8

gis portfolio

## DEFORESTATION MONITORING WEBGIS USING LANDSAT TIME-SERIES AND GOOGLE EARTH ENGINE



crispy-space-xylophone-pjj7g6575pxp29rg6-8000.app.github.dev

**WebGIS Analysis**
Deforestation · Landsat

**Module**
Deforestation
Benchmark period
1990 → 2020

**AOI** Ready
Draw a polygon/rectangle or upload a Shapefile ZIP.
Only the selected module will be executed.
Choose Shapefile ZIP No file chosen
Required in ZIP: .shp + .shx + .dbf + .prj · Max 20 MB
Drawn AOI · AOI loaded.

RUN DEFORESTATION ANALYSIS
CLEAR AOI

Deforestation analysis complete. Only Deforestation was executed.

**Result** Ready
2020 forest 983,607.52 ha
1990 → 2020 -297,441.42 ha change
Forest cover 60.02% of AOI

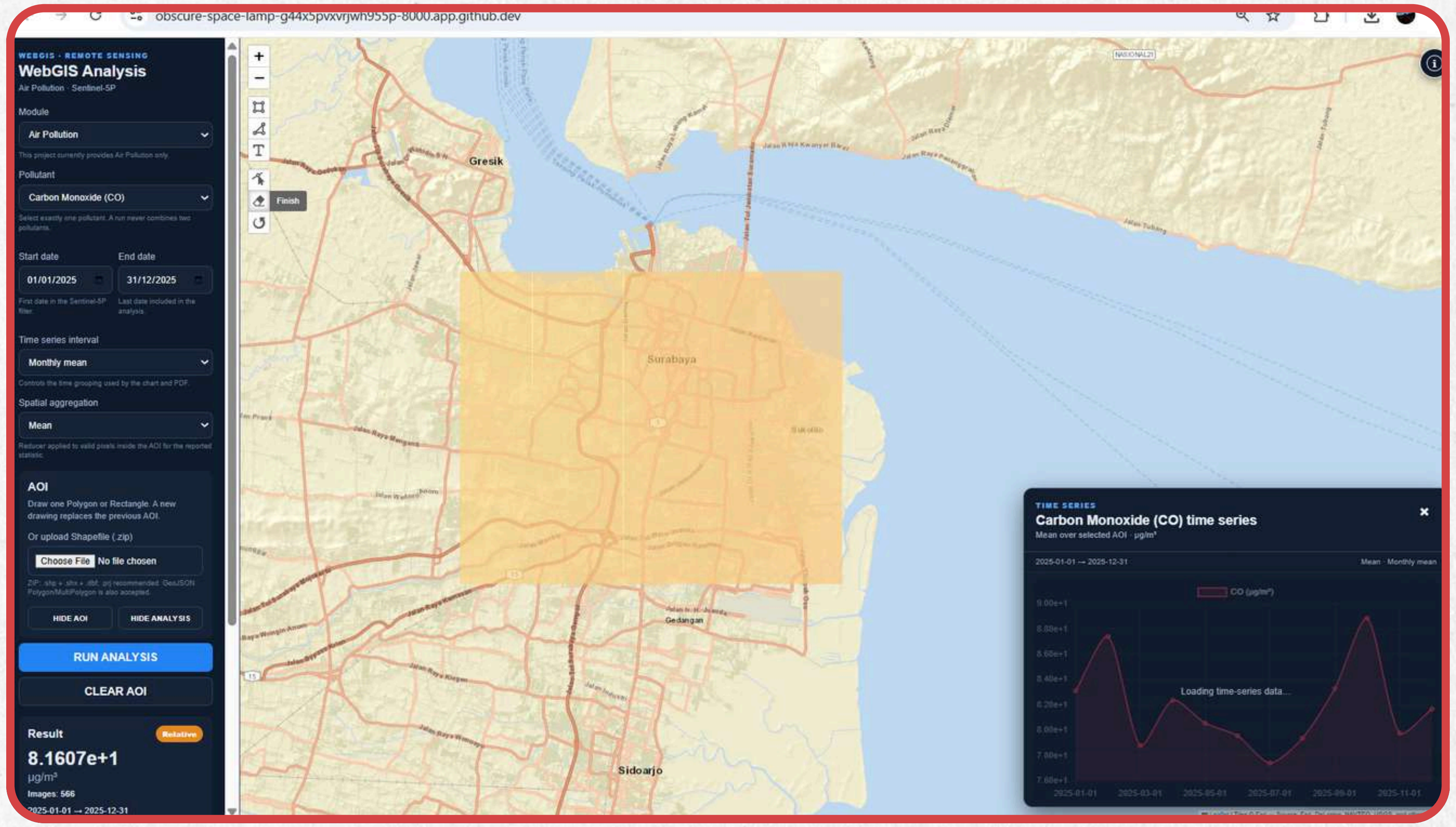
Forest year overlay · use Layers to compare benchmark years

This project develops an interactive WebGIS for monitoring forest cover and deforestation using multi-temporal Landsat imagery and Google Earth Engine (GEE), featuring cloud masking, temporal compositing, forest classification, AOI-based analysis, time-series visualization, and exportable results from 1990–2020. The WebGIS was developed through prompt engineering and AI-assisted coding using GPT-based AI, with the workflow, requirements, and system design directed and refined by the developer.

**Demo**: Currently available as a demonstration through GitHub Codespaces and requires the backend service to be active. The application is not continuously hosted. Please contact me if you would like to test the demo. **Click to Open WebGIS Demo**

# PROJECT 9

## INTERACTIVE AIR POLLUTION WEBGIS USING SENTINEL-5P AND GOOGLE EARTH ENGINE



obscure-space-lamp-g44x5pvxvrjwh955p-8000.app.github.dev
WEBGIS · REMOTE SENSING
Finish
HIDE AOI
HIDE ANALYSIS
RUN ANALYSIS
CLEAR AOI
Relative
Gresik
Surabaya
Gedangan
Sidoarjo
9.00e+1
8.80e+1
8.60e+1
8.40e+1
8.20e+1
8.00e+1
7.80e+1
7.60e+1
2025-01-01
2025-03-01
2025-05-01
2025-07-01
2025-09-01
2025-11-01
CO (µg/m³)
Loading time-series data...

**WebGIS Analysis**

Air Pollution · Sentinel-5P

Module: Air Pollution

This project currently provides Air Pollution only.

Pollutant: Carbon Monoxide (CO)

Select exactly one pollutant. A run never combines two pollutants.

Start date: 01/01/2025

First date in the Sentinel-5P filter.

End date: 31/12/2025

Last date included in the analysis.

Time series interval: Monthly mean

Controls the time grouping used by the chart and PDF.

Spatial aggregation: Mean

Reducer applied to valid pixels inside the AOI for the reported statistic.

AOI

Draw one Polygon or Rectangle. A new drawing replaces the previous AOI.

Or upload Shapefile (.zip)

Choose File No file chosen

ZIP: .shp + .shx + .dbf; .prj recommended. GeoJSON Polygon/MultiPolygon is also accepted.

Result

8.1607e+1

µg/m³

Images: 566

2025-01-01 → 2025-12-31

**Carbon Monoxide (CO) time series**

Mean over selected AOI · µg/m³

2025-01-01 → 2025-12-31

Mean · Monthly mean

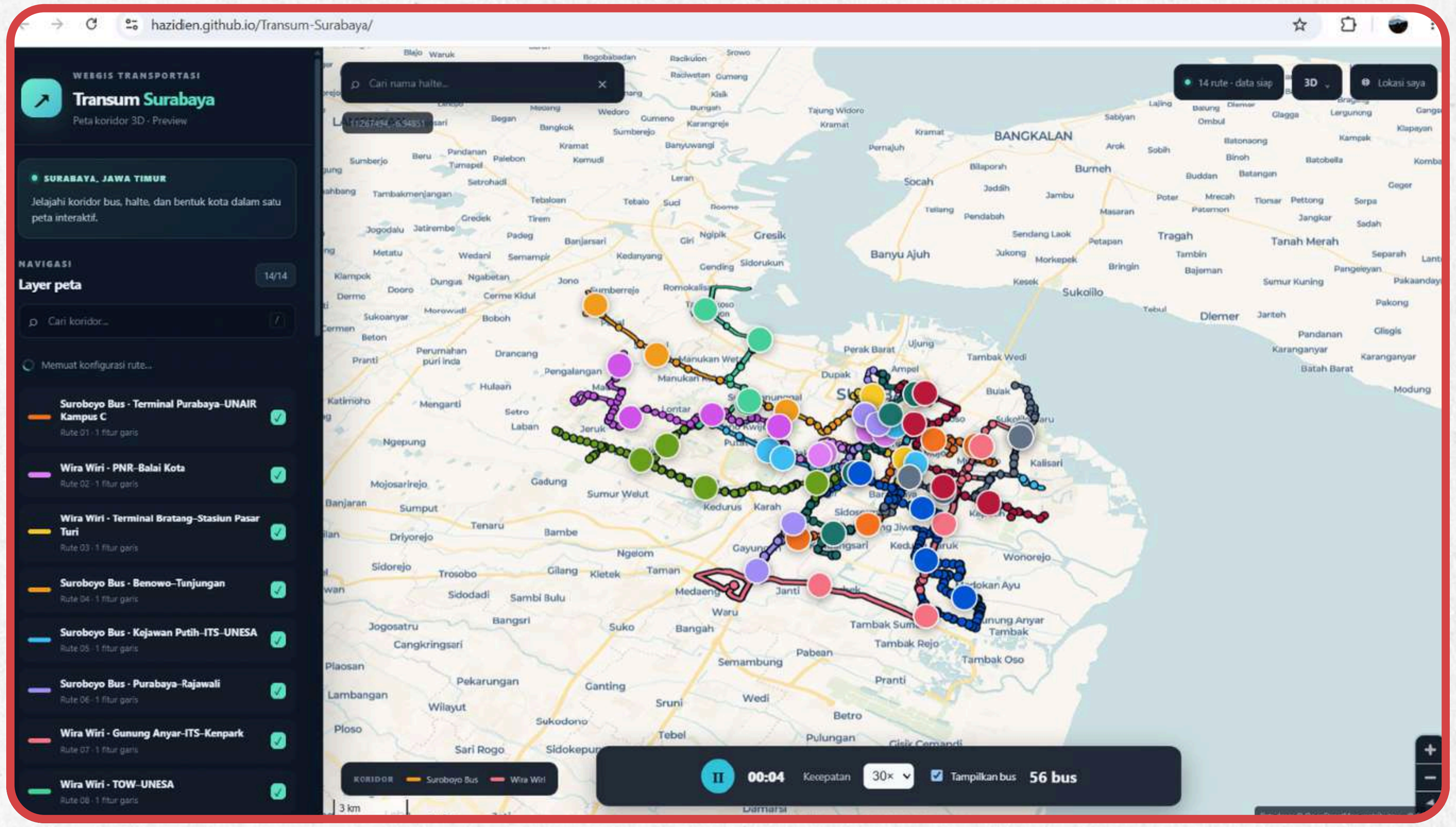
WebGIS for analyzing atmospheric pollution using Sentinel-5P and Google Earth Engine. Users can select pollutants, define dates and AOI, generate spatial concentration maps, statistics, and time-series visualizations. The WebGIS was developed through prompt engineering and AI-assisted coding using GPT-based AI, with the workflow, requirements, and system design directed and refined by the developer.

Demo: Currently available as a demonstration through GitHub Codespaces and requires an active backend service. The application is not continuously hosted. Please contact me if you would like to test the WebGIS.

→ **Click to Open WebGIS Demo**

# PROJECT 10

## 3D WEBGIS FOR PUBLIC TRANSPORTATION ROUTES AND STOP ACCESSIBILITY IN SURABAYA


hazidien.github.io/Transum-Surabaya/
WEBGIS TRANSPORTASI
Transum Surabaya
Peta koridor 3D · Preview
SURABAYA, JAWA TIMUR
Jelajahi koridor bus, halte, dan bentuk kota dalam satu peta interaktif.
NAVIGASI
Layer peta
14/14
Cari koridor...
Memuat konfigurasi rute...
Suroboyo Bus - Terminal Purabaya-UNAIR Kampus C
Rute 01 · 1 fitur garis
Wira Wiri - PNR-Balai Kota
Rute 02 · 1 fitur garis
Wira Wiri - Terminal Bratang-Stasiun Pasar Turi
Rute 03 · 1 fitur garis
Suroboyo Bus - Benowo-Tunjungan
Rute 04 · 1 fitur garis
Suroboyo Bus - Kojawan Putih-ITS-UNESA
Rute 05 · 1 fitur garis
Suroboyo Bus - Purabaya-Rajawali
Rute 06 · 1 fitur garis
Wira Wiri - Gunung Anyar-ITS-Kenpark
Rute 07 · 1 fitur garis
Wira Wiri - TOW-UNESA
Rute 08 · 1 fitur garis
Cari nama halte...
14 rute · data siap
3D
Lokasi saya
BANGKALAN
KORIDOR
Suroboyo Bus
Wira Wiri
00:04
Kecepatan
30×
Tampilkan bus
56 bus
3 km

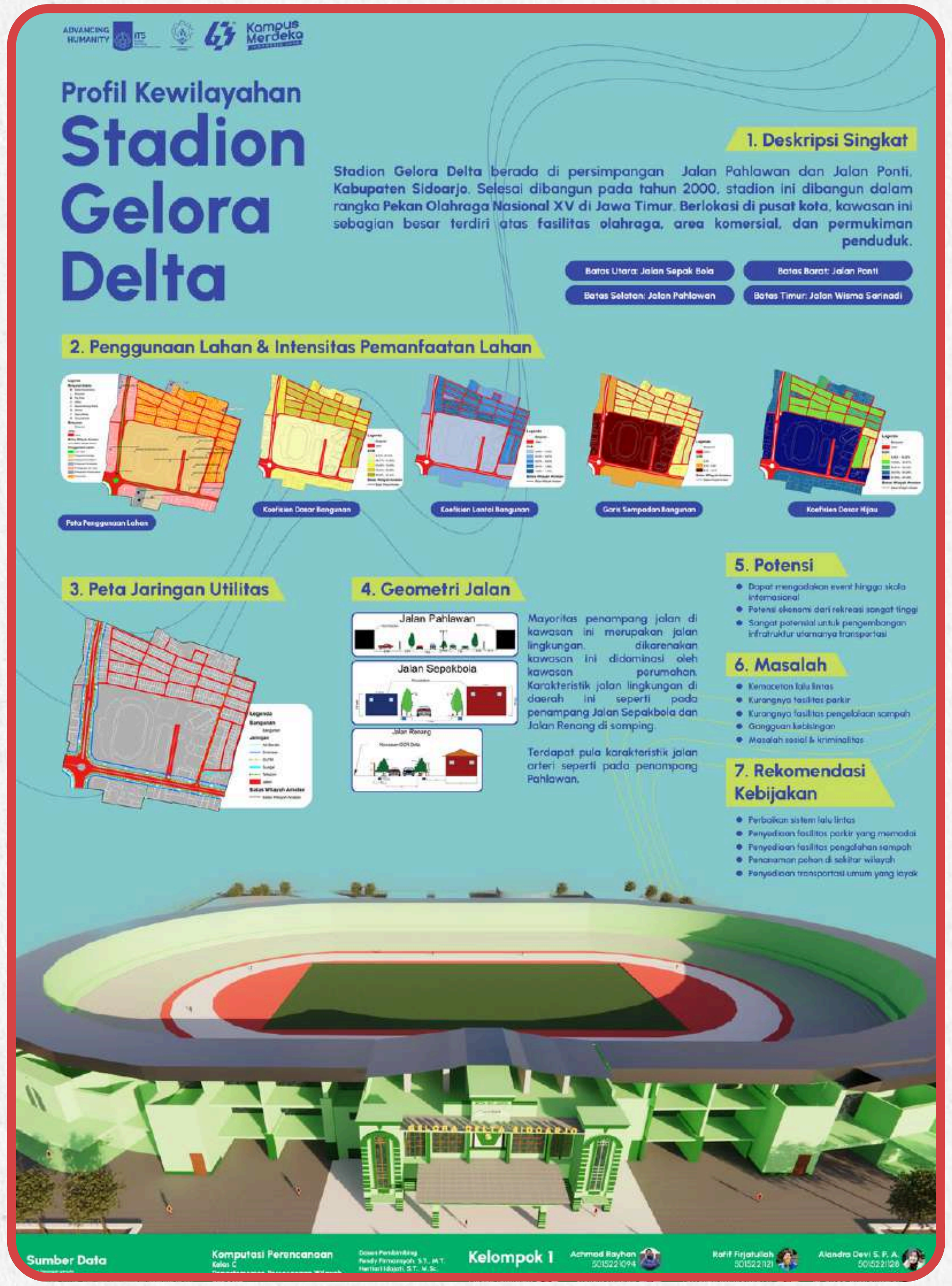
Developed an interactive 3D WebGIS featuring 14 Suroboyo Bus and Wira Wiri routes, including route networks and bus stop locations across Surabaya. Integrated interactive stop and route information to display the number of buses serving each stop, total stops per route, and route length (km). Implemented spatial accessibility analysis to identify the nearest bus stops, estimate walking distances and travel times from user-selected locations, and visualize adjustable stop coverage buffers ranging from 100 m to 1 km.

→ **Click to Acces**

# PROJECT 11

## URBAN SPATIAL ANALYSIS AND BUILDING DEVELOPMENT PLANNING OF GELORA DELTA STADIUM, SIDOARJO

Conducted a comprehensive spatial analysis of the Gelora Delta Stadium area, Sidoarjo, focusing on land use patterns, building intensity, and urban development. Assessed building coverage ratio (KDB), floor area ratio (KLB), green area ratio (KDH), and building setback lines (GSB). Evaluated building massing and area development patterns, land use, utility networks, and road geometry to identify spatial potentials, infrastructure constraints, and urban issues. Developed planning recommendations and a 3D stadium model to support integrated spatial planning and area development.



Profil Kewilayahan

**Stadion Gelora Delta**

1. Deskripsi Singkat

Stadion Gelora Delta berada di persimpangan Jalan Pahlawan dan Jalan Ponti, Kabupaten Sidoarjo. Selesai dibangun pada tahun 2000, stadion ini dibangun dalam rangka Pekan Olahraga Nasional XV di Jawa Timur. Berlokasi di pusat kota, kawasan ini sebagian besar terdiri atas fasilitas olahraga, area komersial, dan permukiman penduduk.

Batas Utara: Jalan Sepak Bola

Batas Barat: Jalan Ponti

Batas Selatan: Jalan Pahlawan

Batas Timur: Jalan Wisma Sarinadi

2. Penggunaan Lahan & Intensitas Pemanfaatan Lahan

3. Peta Jaringan Utilitas

4. Geometri Jalan

Mayoritas penampang jalan di kawasan ini merupakan jalan lingkungan, dikarenakan kawasan ini didominasi oleh kawasan perumahan. Karakteristik jalan lingkungan di daerah ini seperti pada penampang Jalan Sepakbola dan Jalan Renang di samping.

Terdapat pula karakteristik jalan arteri seperti pada penampang Pahlawan.

5. Potensi

- Dapat mengadakan event hingga skala internasional
- Potensi ekonomi dari rekreasi sangat tinggi
- Sangat potensial untuk pengembangan infrastruktur utamanya transportasi

6. Masalah

- Kemacetan lalu lintas
- Kurangnya fasilitas parkir
- Kurangnya fasilitas pengelolaan sampah
- Gangguan kebisingan
- Masalah sosial & kriminalitas

7. Rekomendasi Kebijakan

- Perbaikan sistem lalu lintas
- Penyediaan fasilitas parkir yang memadai
- Penyediaan fasilitas pengolahan sampah
- Penanaman pohon di sekitar wilayah
- Penyediaan transportasi umum yang layak

Sumber Data

Komputasi Perencanaan
Kelas C

Dosen Pembimbing
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Hertiari Idajati, S.T., M.Sc.

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LET'S WORK

# TOGETHER!

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